WEATHER AND INCOME'S IMPACT ON CAPITAL BIKESHARE RIDES

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BACKGROUND

IMPORTANCE

- Bikesharing is a popular, low-cost method of transportation in cities
- Transportation is critical
- Easy and cheap transportation methods should be accessible to those who need it the most
- If weather disproportionately affects lower income neighborhoods, transportation eauity could be affected

RESEARCH QUESTION

Does weather have a more extreme effect on Capital Bikeshare ridership in lower income neighborhoods in DC?

DATA

- Capital Bikeshare Data
- Nation Oceanic and Atmospheric Administration (NOAA)
  - DCA Weather Station
- Census Income Data

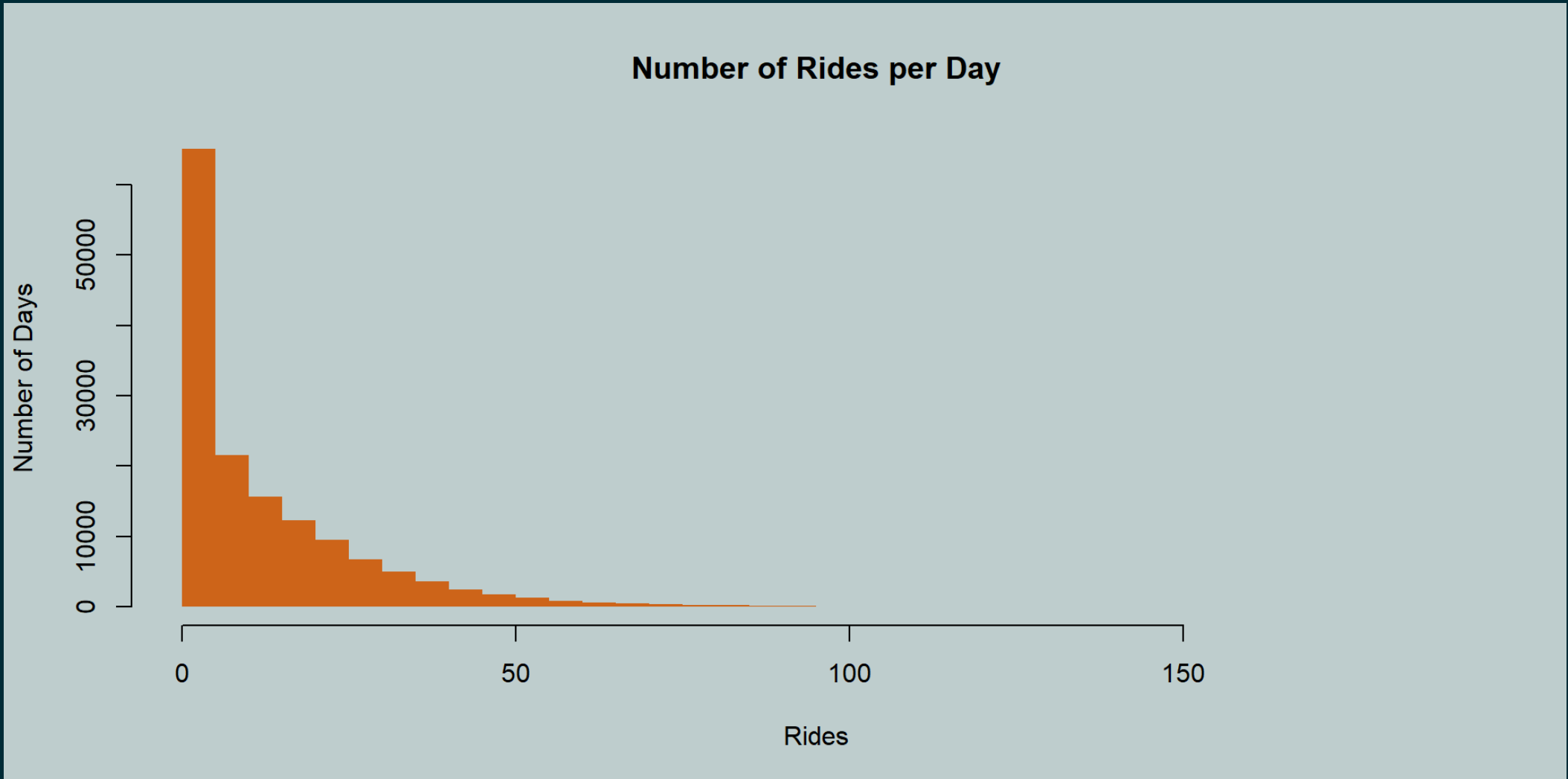
CHALLENGES AND LIMITATIONS WITH THE DATA

- Missing values
  - Removing e-bikes
  - DCA weather station
  - Imputation for wind speed
- Rides outside DC boundaries were removed

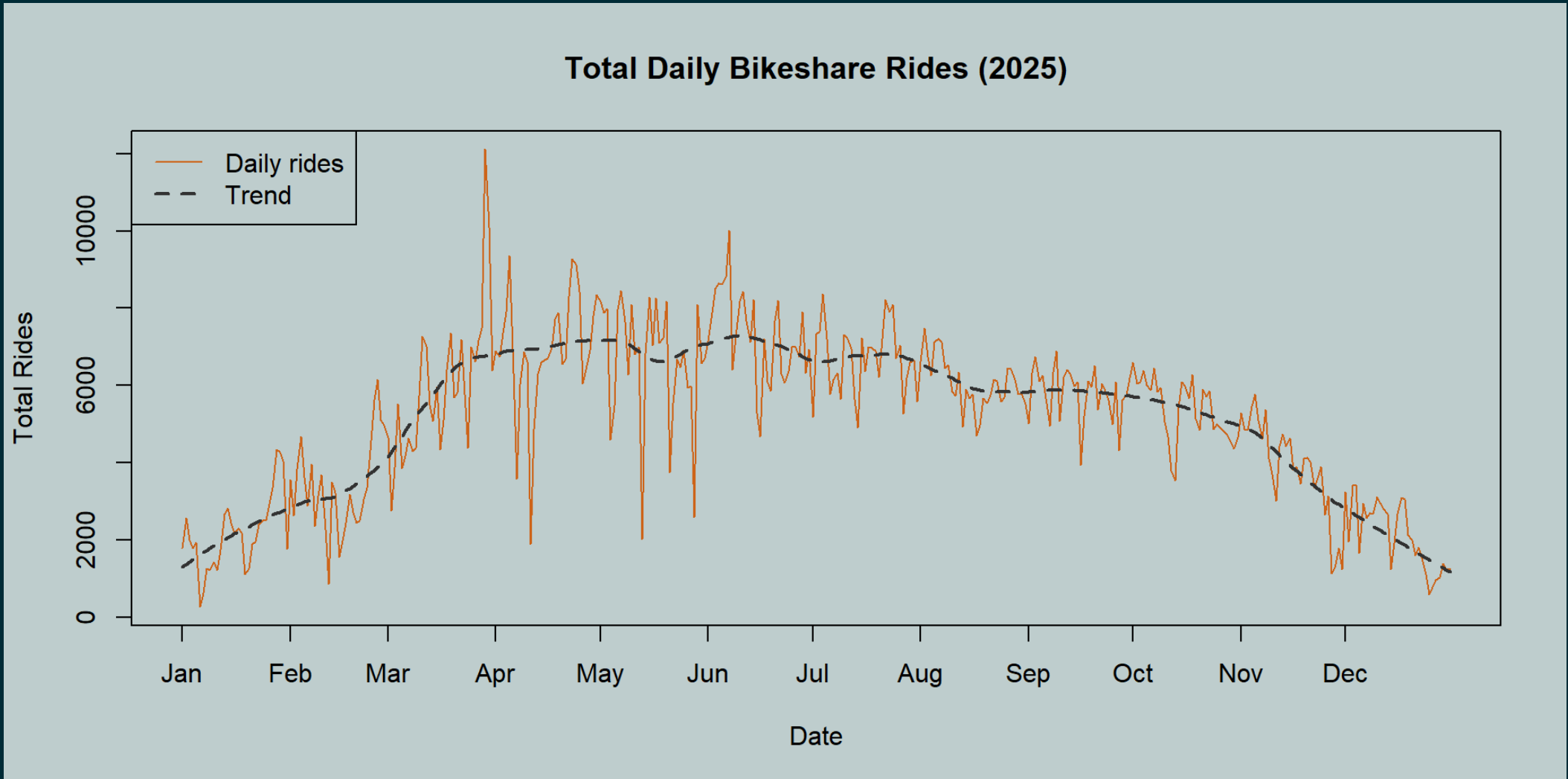
DATA STRUCTURE

Station	Date	Rides	Med. Income	Rain (in.)	Snow (in.)	Mean Temp. (°F)
9th St & Pennsylvania Ave NW	2025-01-01	4	151659	0.00	0.0	46.0
9th St & Pennsylvania Ave NW	2025-01-02	5	151659	0.00	0.0	41.0
9th St & Pennsylvania Ave NW	2025-01-03	2	151659	0.04	0.0	38.5
9th St & Pennsylvania Ave NW	2025-01-04	3	151659	0.00	0.0	30.0
9th St & Pennsylvania Ave NW	2025-01-05	1	151659	0.00	0.0	30.5
9th St & Pennsylvania Ave NW	2025-01-06	0	151659	0.81	7.2	31.0

DAILY RIDES


Number of Rides per Day
Number of Days
0
10000
30000
50000
0
50
100
150
Rides

TIME SERIES


Total Daily Bikeshare Rides (2025)
Daily rides
Trend
Total Rides
0
2000
6000
10000
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Date

MODELING

GOALS

- Want to model a daily ride count using weather, income, and other socioeconomic predictors
  - Poisson model
- Does the effect of weather vary by neighborhood income?
  - Weather x income interactions

POISSON

- Assumes:
  - data follows Poisson distribution
  - mean = variance

MODEL HERE?

explain Poisson model..maybe use Latex to write out on another slide

POISSON RESULTS

- Overdispersion
- Assumptions clearly violated table

NEGATIVE BINOMIAL

- Poisson assumption (mean = variance) was badly violated
- Poisson model dispersion statistic = 14.21
  - Variance is nearly 12× the mean
  - Standard errors become too small, making results look more significant than they are
- Negative Binomial model adds a parameter to absorb extra variability
- Dispersion drops to 1.17

NEGATIVE BINOMIAL

MODEL

explain the negative binomial model, the full, then the simplified

COMPARISON

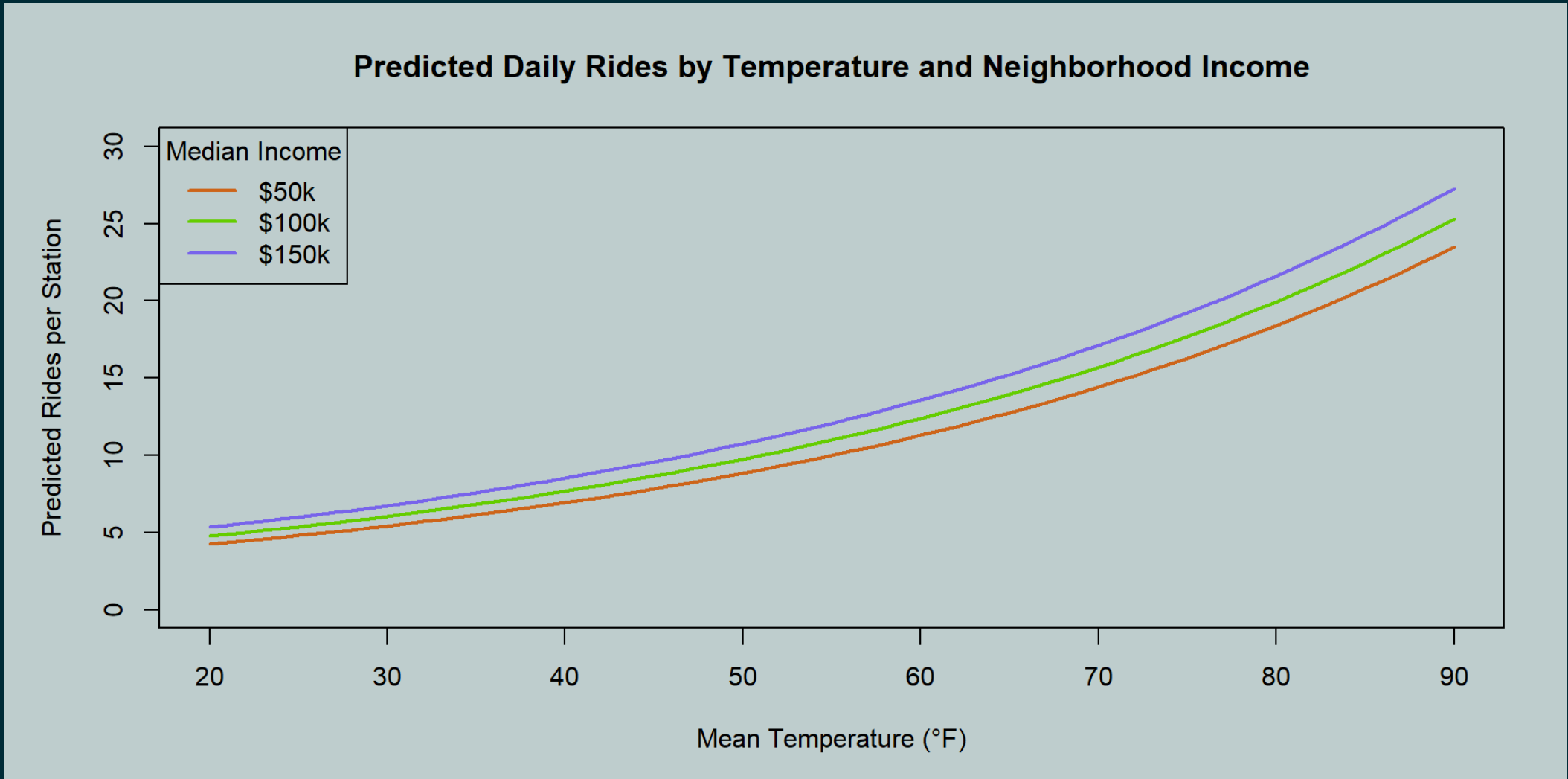
Model	AIC	Dispersion	RMSE
Poisson	2596830	14.21	14.81
Negative Binomial	1025364	1.17	14.94

INTERPRETATION

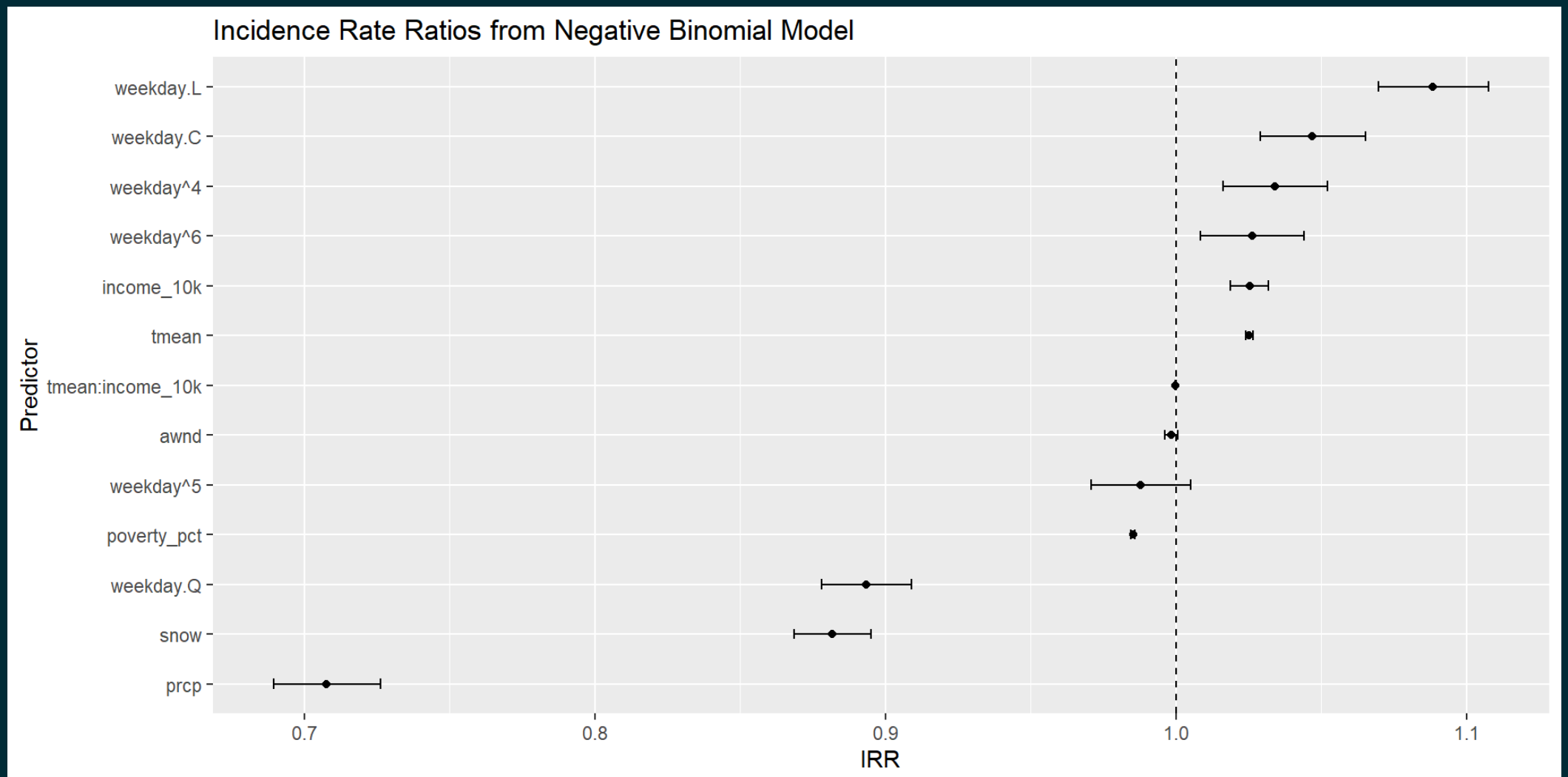
INTERACTION TERM

Term	IRR	CI Low	CI High	p-value	
Intercept	2.686	2.464	2.929	0.000	***
Temperature	1.025	1.024	1.027	0.000	***
Precipitation	0.707	0.689	0.726	0.000	***
Snow	0.882	0.869	0.895	0.000	***
Wind	0.998	0.996	1.001	0.120	
Income ($10k)	1.025	1.019	1.032	0.000	***
Poverty %	0.985	0.984	0.986	0.000	***
Weekday.L	1.088	1.070	1.108	0.000	***
Weekday.Q	0.893	0.878	0.909	0.000	***
Weekday.C	1.047	1.029	1.065	0.000	***
Weekday^4	1.034	1.016	1.052	0.000	***
Weekday^5	0.988	0.971	1.005	0.132	
Weekday^6	1.026	1.009	1.044	0.001	**
Temp × Income	**1.000**	**1.000**	**1.000**	**0.002**	**

INCOME X WEATHER INTERACTION



COEFFICIENT PLOT



CONCLUSIONS

- The interaction has a small, but significant impact on the count of rides.
- other conclusions

THANK YOU